CO3202 Entrepreneurial Project 2023-24

INTERIM REPORT

Youssef Sadquy – Career Fair in Unity

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INTRODUCTION

The fast evolving world and digitalisation of daily life aspects has influenced the creation of "GlobeFair". It was born from the idea of creating a powerful software to transform the traditional career fair into a more entertaining virtual experience. The project will make use of Unity to redefine the different aspects of networking and career opportunities for a global audience. The following report will concentrate on presenting the aim, objectives, and challenges for GlobeFair, followed by a background research and development of strategies with a business approach, pursuing then into the software requirements, technical aspects, and planning of the project.

1. AIMS AND OBJECTIVES

1.1 Aims

The project core aim is to develop a software capable of recreating the atmosphere of an in-person career fair event, while being entirely virtualised, more accessible, cost effective, scalable, targeting a broader audience, and offering more tools to jobseekers.

1.2 Objectives

To achieve the project aim, the project will pursue the following objectives:

1. **Market Analysis:** Conduct extensive research to understand existing platforms and identify unique opportunities and gaps in the market.
2. **Software Design:** Develop a detailed plan for the software, defining features and user interfaces.
3. **Unity Development:** Create a completely functional virtual environment in Unity that simulates the experience of an in-person career fair.
4. **User Interaction Tools:** Integrate features for live interactions, such as chat systems, multimedia presentations, virtual booths, and file exchanging.
5. **Data Management:** Develop a secure database to handle backend operations like user data storage and privacy.
6. **Global Accessibility:** Ensure the platform is accessible and inclusive for users worldwide, with a focus on user friendly design.

1.3 Challenges

The project also presents some challenges:

- **Digital Transformation:** Innovating in the digital space to replicate and enhance the traditional career fair experience.
- **User Experience and Engagement:** Designing an intuitive and engaging platform that successfully simulates in person interactions in a virtual environment.
- **Data Security and Privacy:** Ensuring robust data protection measures to safeguard user information.
- **Scalability and Adaptability:** Building a platform capable of accommodating various event sizes and formats, from small circle networking sessions to larger scale career fairs.

2. SURVEY OF LITERATURE/INFORMATION SOURCES

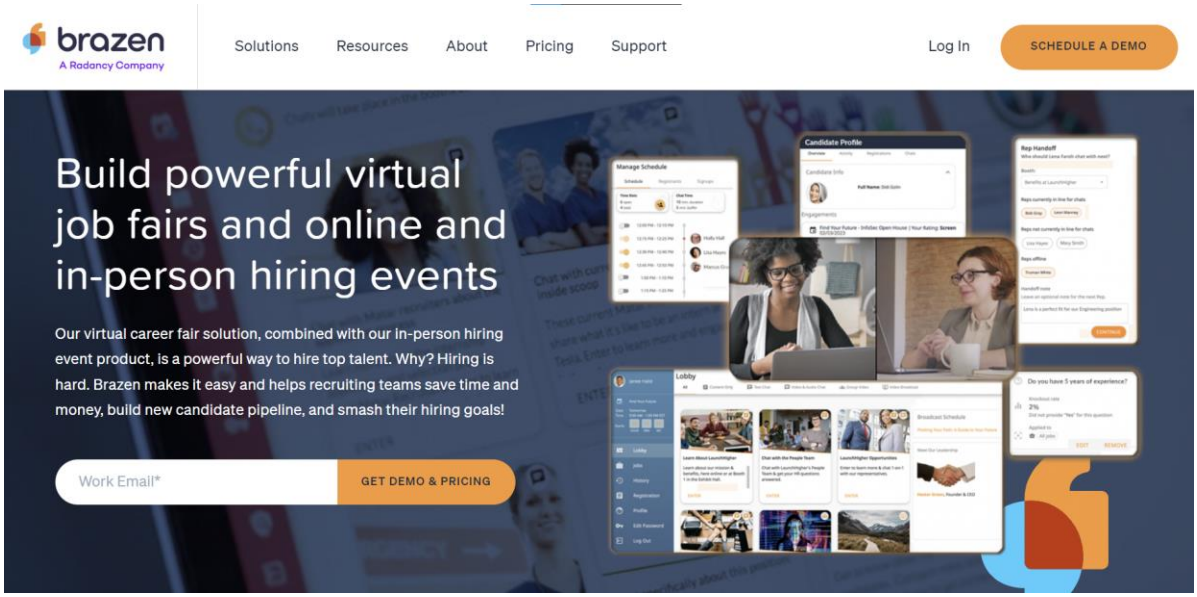
2.1 Target Audience Research

The focus on students and young professionals as the primary audience for GlobeFair aligns with extensive research into the shaping employment market methods and jobseekers preferences. Analysis emphasizes the growing significance of online talent platforms, projecting a potential contribution of $2.7 trillion to global GDP by 2025, thus enhancing employment market efficiency and transparency[1]. This trend resonates with the digital trends of young job seekers, as evidenced by the research findings, which reveals a substantial reliance, by students and graduates, on internet platforms for job hunting[2]. The COVID-19 pandemic has increased this digital shift, which highlights the necessity for innovative, accessible virtual career platforms, particularly suited to the digitally skilled demographics which are students and young professionals seeking career development and networking opportunities in a fast paced digital world[3].

2.2 Existing Virtual Platforms Research

In researching existing virtual event platforms, my evaluation was centred on Brazen.com[4], a platform recognised for its efficiency in the virtual hiring landscape. It offers a combination of virtual, in person, and hybrid event scenarios. It offers features for example automation, workflow optimization, and integrated communication tools such as text, audio, and video. Figure 2.1 illustrates the core structure of Brazen[4].

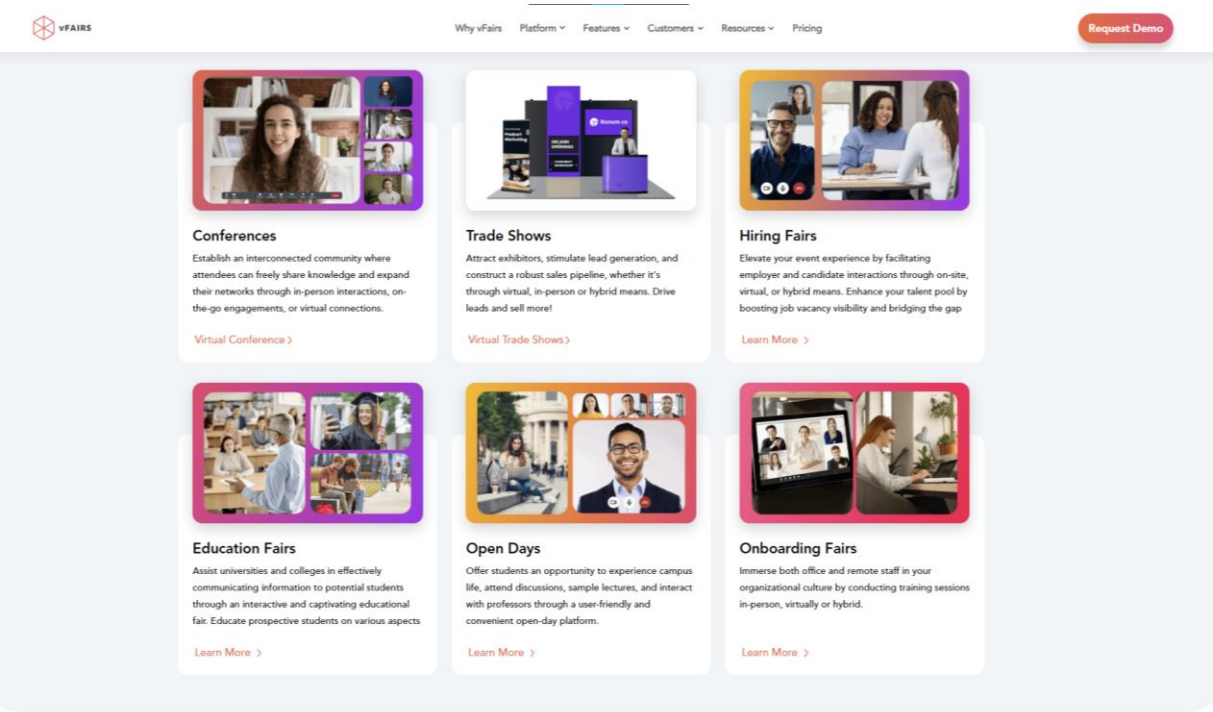
Figure 2.1



However, the analysis uncovered gaps in existing platforms like vFairs[5] and 6Connex[6], including Brazen, which GlobeFair aims to address. Current virtual career platforms do not entirely leverage immersive visualization, which can significantly enhance the virtual experience. They also lack skill based matchmaking, a crucial feature for aligning candidate capabilities with employer requirements. Furthermore, while platforms like Brazen offer effective recruitment tools, they overlook post fair engagement processes essential for supported communication and candidate development.

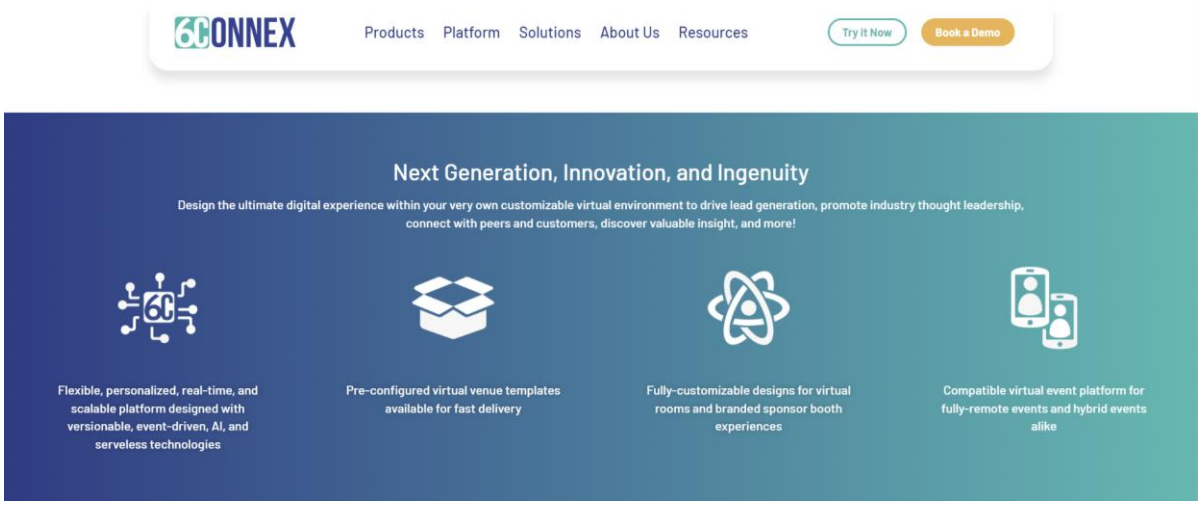
Figure 2.2 illustrates the core structure of vFairs[5] Similarly to Brazen the platforms offers in person, hybrid, and virtual environment fairs. It offers limited post event engagement functionalities, which are deemed prominent in the GlobeFair vision to help the jobseeker in developing their career post fair and maintaining a relationship with the recruiters.

Figure 2.2



Additionally, on figure 2.3, 6Connex[6] does not offer features like real-time language, limiting accessibility for a global audience. The absence of specific engagement tools, such as tangible rewards, is another area where current platforms lack. These elements are vital for enhancing user interaction and engagement, making the virtual career fair experience more dynamic and interactive.

Figure 2.3

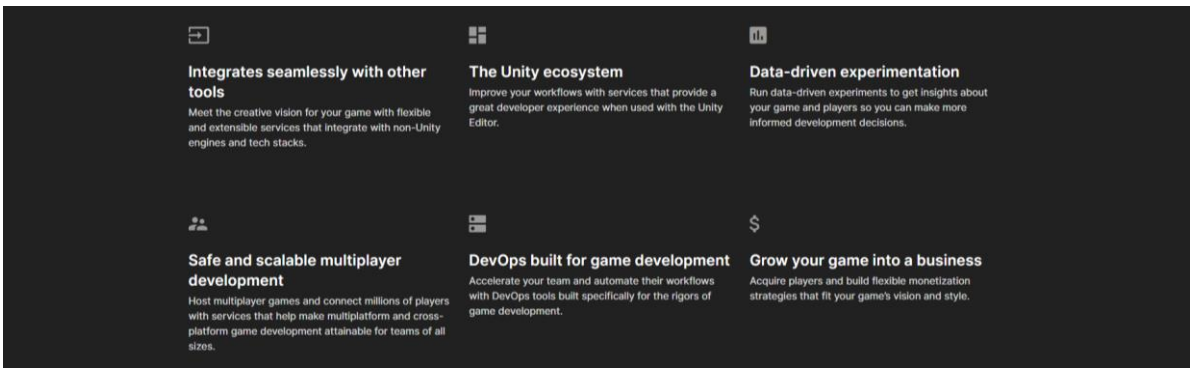


In addressing these market gaps, the proposed software will not only offer the standard benefits of virtual hiring events, such as cost savings, increased talent reach, but also improve the experience with innovative features. Enhanced data tracking and analytics for both jobseekers and recruiters, will provide deeper insights into candidates behaviour and event success, ensuring a constantly improving model that goes beyond the current market offerings.

2.3 Technology Research

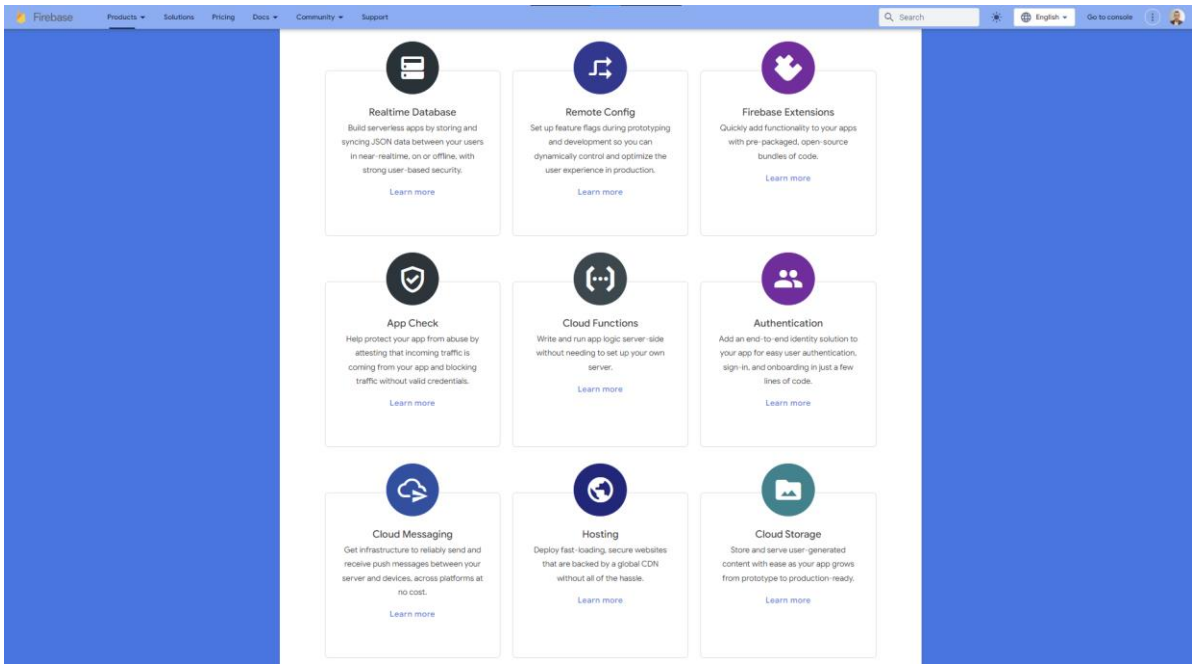
Technologies have to be researched to guarantee a successful implementation of GlobeFair. Unity game engine, chosen for, the cross platform functionalities it offers, ease of use in designing/developing 3D interactive environments and the vast amount of documentation it offers to the developer[7] as illustrated in figure 2.4, makes it a key player for the innovations that GlobeFair aims to implement. Unity's also offers render pipelines, such as the High Definition Render Pipeline (HDRP)[8] and Universal Render Pipeline (URP)[9], such that it allows for flexibility in graphical rendering, which is crucial for creating the immersive virtual environment for this project.

Figure 2.4



During the research for the appropriate backend technology, Firebase, a Backend-as-a-Service(Baas) offered by Google[10], seemed like the most appropriate solution to bring GlobeFair vision to life. Firebase Suite does not only offer database functionalities, but real time database which makes it a prime candidate for GlobeFair purpose to bring people together, additionally offers authentication services, ensuring that users data is encrypted and safe from external attacks. Moreover, cloud storage, cloud functions and hosting services it offers could easily help the project scale to a global audience hassle free, as represented in figure 2.5.

Figure 2.5



2.4 Unique Features Research

The project concentrated on the research of unique features to enhance the current market offerings, the discovery of real time language translation for the user chat, is enabled thanks to the Google Cloud Translation API[11] which will be integrated to Vivox, which offers real time cross platform chat functionalities[12], the research highlighted the lack of matchmaking engine and skill assessment for jobseekers to facilitate the job hunting process. Additionally, post fair engagement data, as mentioned above, will facilitate continuous communication and interaction with recruiters with the ability to integrate the Firestore system for CV uploads and submission at booths.

3. BUSINESS DEVELOPMENT

3.1 - Unique Value Proposition (UVP)

"Virtualising Global Career Networking", is GlobeFair's UVP, it captures in few words the aim of the software to allow its users to connect, network and improve their career settings in a virtualised environment and digital experience. This improves the aim of GlobeFair by creating an impact and to transform the traditional career fair into a more immersive reality, breaking down geographical barriers, not just by offering an accessible solution to the current market but more engaging and cost effective for young professionals. This approach has been proven effective in gaining popularity for different platform backgrounds[13].

3.2 – The Customers

GlobeFair's customer base includes educational institutions, corporations, and student/recent graduates. The customer development process involved market research, by studying the currently existing platforms, I identified the gaps, and finally gathered actual feedback from the customer base by completing a survey, as shown below in appendix 8, figure 8.1.

The survey, which has been sent to 20 jobseekers using Google Forms to answer multiple questions on career fair and their current employment status, has given significant results to conduct the following analysis:

Demographic Profile:

The majority of the responses revealed that 55% were currently in studies, while 35% of the responses indicated that 35% have recently graduated(within the past 2 years) and do not have a job yet. This data aligns with GlobeFair aimed customer base, which additionally highlights the importance of young people, students, and recent graduates as the core part in the jobseekers area within the digital space. Appendix 8, Figure 8.2.

Career Fair Preferences:

To the next question, data showed that 60% of the responses were for the preference of virtual career fairs rather than in person ones. This also a key indicator as the majority of the respondents aligns with GlobeFair purpose, additionally it highlights the growing digital trend by proving the convenience of having the same experience, with additional features, wherever the person is located. Appendix 8, Figure 8.3.

Importance of Advanced Features:

The most interesting factor about the virtual career fair, as highlighted by the survey is the presence of advanced features, such as interactive booths, real time chat and CV uploading and submission, these features appealed to the respondents, 80% of the responses showed the importance of

including these features to enhance the user overall experience as well as having more engaging activities. Appendix 8, Figure 8.4.

Frequency of Traditional Fair Attendance:

The survey also centred on the actual attendance of the participants; the data showed that only 40% attended “Often” or “Very Often”, while the rest has either never participated or didn’t go as frequently, this data suggests a shift towards online platforms for communication and job hunting processes which has become more accessible, more convenient, and more beneficial in terms of features and value in return for the user. Appendix 8, Figure 8.5.

Satisfaction with Traditional Fairs:

The final question reviewed the overall satisfaction of the users with the in person fair, this if they have participated, 50% of the participants responded in a positive way and they were satisfied with the current offerings, while the rest had mixed feeling between negativity and neutrality, appendix 8, figure 8.6, this data is significant as it shows there’s room for GlobeFair to address the common satisfaction issues faced by the survey participants, approximately 70% of respondents expressed a desire to have more engaging activities, and to have more networking opportunities, often these events lack time to discover in their entirety or are too crowded. Appendix 8, Figure 8.7.

Conclusion and Market Potential:

In conclusion the survey has presented strong data which proves GlobeFair’s good strategy for the selection for the customer base, mainly students and recent graduates, having 60% of the participant preferring a virtual setting, 80% valuing more advanced features than the ones currently available, there is therefore an overall favour for innovation and virtualisation of the career fair by the participants which aligns with the GlobeFair’s aim and the evolving career needs.

3.3 - Market Strategy

3.3.1 - Lean Canvas

The lean canvas presented below, figure 3.1, offers an overview of GlobeFair business model. The lean canvas is a fundamental tool that allows to quickly visualise the key elements of the business structure and validate the ideas and concepts[14], such as the main problems, the solutions, customer segments, cost structure and revenue strategies.

Figure 3.1

Problem	Solution	UVP	Unfair advantage	Customer segments
• Lack of immersive visualisation • Lack of Skill-based matchmaking • Difficulty in sharing CVs at virtual fairs • Lack of post-fair engagement • Users struggle to track their interactions • Unhandled language barriers	• Immersive virtualisation in Unity • Skill-based matchmaking • Real-time translation tools • CV upload and submission directly to company booths • End of session reports for both students and companies	"Virtualising Global Career Networking"	• Post-event engagement and skill-based matchmaking • Real-time translation integrated	• Students and young professionals • Educational institutions and corporations
Existing alternatives	**Key metrics**		**Channels**	
• Brazen.com • VFairs.com • 6Connex.com	• Number of fairs hosted • Number of students and companies registered • CVs submitted per fair • Active user engagement time		• Online marketing • Partnerships with educational institutions • Webinars and promotional events	
Cost structure			**Revenue streams**	
• Development (platform, translation tools, backend infrastructure) • Marketing and promotion • Server and hosting costs • Partnerships and affiliations • Maintenance and support			• Subscription models for institutions and companies • Partnership and sponsorship deals for fairs • Companies pay for premium visibility on a highlighted job posting board • Licensing the software for private use by larger institutions or corporations	

3.3.2 - Understanding the Competitive Environment

GlobeFair adopted a competitive approach by targeting directly existing market gaps, it differs from the competition as it offers to the users innovative features such as immersive visualisation, matchmaking engine for quick user and recruiters match, real time chat with other users and recruiters and the crucial post fair engagement dashboard. The approach has been adopted after extensive market research and by analysing the strengths and weaknesses of multiple virtual career fair platforms[4][5][6].

3.4 - Revenue Model and Growth Strategy

GlobeFair revenue model aims to offer to its customer base different types of paid services, these include subscription fees, partnership deals and premium visibility options for the recruiters. The subscription model allows to have a continuous revenue stream from educational institutions and businesses, while partnership deals to gain popularity amongst the competitors and to establish the brand, premium visibility, while not being always steady, guarantees extra revenue from each event.

For GlobeFair financial estimation for the first year, is designed to build brand awareness while generating revenue. An estimation for the first year is of approximately £150000, as shown on the breakdown below:

1. Subscription fees are estimated to bring revenue for £50000, by charging institutions and businesses on a per event basis a fee of £500. The estimation is to host 100 events on the first year or 2 per week.
2. While partnership deals, will involve industry collaborations or educational partners to licence the software, the aim is to secure 10 partnerships on the first year, each funding £5000, securing revenue for £50000.
3. Premium visibility options for recruiters will be charged at the rate of £200 per event, on the course of the first year the target is of 250 recruiters or 2.5 per event, generating a revenue of £50000.

Regarding the expenses estimations, the initial year is predicted to see expenses of around £110000, as shown on the breakdown below:

1. The initial development costs are estimated to be £10000, this covers the software development, the software testing and the initial setup cost based on the amount of users.
2. App deployment is estimated to cost £30000 and will include flexible hosting services to accommodate different event sizes, to ensure scalability, additionally it will include professional testing from third parties to guarantee the robustness of the software.
3. Marketing and Operations costs will total at approximately £30000, which will be used for marketing on different social media and web engines, to establish the brand and to hire staff.
4. Finally, there will be customer support and maintenance costs of £40000 to ensure the software is always up to date and the customer requests are addressed.

The focus will be on reinvesting the initial profits for enhancing the software, offering more features to its customer base, expansion, and long term growth.

3.5 - Innovative Business Development Initiative

The strategic partnership with educational institutions aims to integrate directly with the institution's career services, this will allow students to have a smooth transition from education to employment. This innovation will enhance user engagement as well as build brand awareness in the career networking space.

4. REQUIREMENTS

4.1 - Must-Haves

Application Development:

- Unity Integration: Implement the core application using Unity to provide an immersive 3D experience, ensuring it can handle multiple users simultaneously with robust performance.
- API Integration: Incorporate APIs for real time translation, CV uploads, and matchmaking algorithm to enhance user interaction and data processing capabilities.
- Stability and Scalability: Architect the system to ensure stability under high traffic and scalability for future growth.

Account Features:

- Sign Up/Log In: Develop a secure authentication system with encryption and best practices for data protection.
- Profile Creation: Create a dashboard for profile management system that allows for detailed user information input and editing.
- Password Recovery: Implement a secure and user friendly password recovery process.

Database Features:

- User Data Storage: Integrate encrypted databases to store sensitive user information securely.
- Company Data Storage: Safeguard company information, booth designs, and job postings with regular security assessments.
- Analytics Storage: Collect and store engagement metrics with privacy compliance and real time data analysis.

Interface Features:

- Interactive 3D Environment: Design an intuitive, immersive, and engaging 3D space that simulates the physical career fair environment.
- Navigation Controls: Develop user friendly navigation controls adaptable to various user interaction modes.
- Responsive Design: Ensure cross device compatibility and responsiveness.

Student Features:

- Avatar Customization: Integrate a flexible avatar customization tool.

- CV Upload: Implement a CV management system with the ability for users to upload and manage their documents.
- Booth Interaction: Create an interactive system for students to engage with company booths.
- Summary Metrics: Provide analytics on user engagement post event.

Companies Features:

- Booth Personalization: Develop a booth customization toolkit for companies.
- Job Posting: Create a job management system allowing companies to post and manage job listings.
- Receive CVs: Build a CV collection and management system for company recruiters.
- Metrics Dashboard: Offer a complete analytics dashboard for companies to review post event engagement data.

4.2 - Preferable Features

Application Development:

- Cross-platform Compatibility: Ensure the application runs smoothly on various platforms.
- Light and Dark Mode: Implement a theme system based on user preference.

Account Features:

- Social Media Integration: Integrate with social media platforms for easier account creation or logging in.
- Account Verification: Establish a verification system for company accounts.

Database Features:

- Backup & Recovery: Set up automated and manual backup and recovery methods.
- Search Functionality: Develop an advanced comapny search engine within the platform for the users.

Interface Features:

- Interactive Tutorials: Create tutorials to assist new users in navigating the platform.
- Feedback Mechanism: Implement a system that allows to collect user feedback and bug reports.

Student Features:

- Bookmarking: Allow students to bookmark their favourite companies.
- Networking Spaces: Facilitate networking through dedicated virtual spaces.

Companies Features:

- Live Webinars/Workshops: Enable companies to host live sessions in dedicated virtual spaces.
- Interactive Media: Allow companies to integrate various media types into their booths.

- Analytics Export: Provide data export functionality in multiple formats.

Additional Features:

- Real Time Chat: Implement real time chat functionality for instant communication.

4.3 - Optional Features

Application Development

- Voice Commands: Integrate voice navigation abilities.

Account Features:

- Achievement Badges: Implement a badge/reward system for user engagement.
- Two-Factor Authentication: Add an extra security layer for user accounts.

Database Features:

- Recommendation Engine: Develop an algorithm to suggest companies or roles to students.
- Tagging System: Allow companies to tag their listings for easier discovery.

Interface Features:

- Personalized Dashboards: Create customizable dashboards for users.

Student Features:

- Interactive CV Builder: Provide tools for creating and formatting CVs.
- Virtual Business Cards: Enable the creation and sharing of digital business cards.

Companies Features:

- Q&A Forums: Set up forums for company and student interaction.
- Virtual Goodie Bags: Offer digital content packages for booth visitors.

Additional Features:

- API Integrations: Allow for third-party integrations to extend platform capabilities.

5. Outline of Specification and Design

5.1 – System Architecture

The core engine that will be used for the implementation of GlobeFair will be Unity, the software will be combined with the Model View Controller(MVC) pattern. The main programming language will be C#, which will allow strong memory management and a set of features for complex software logic, for example the real time 3D rendering of the virtual environment. The MVC patter, fundamental aspect of this project, will allow a well structured code, divided into user interface(View), business logic(Model) and control flow(Controller), allowing easier maintainability and scalability[15].

5.1.1 – Technology Stack

- Unity and MVC: Unity has been selected, for its wide functionalities in the creation of real time 3D interactive environments using C#. While the MVC pattern will further enhance the code structure by applying a clear separation of different activities especially for more advanced/complex features within the environment.
- Firebase Suite: It has been selected for its vast number of functionalities, the core features that will be used is the real time database, which allows simultaneous connectivity of multiple users and interaction, it is essential for the project to recreate the atmosphere found in traditional career fairs[16]. Firestore, which is part of the suite, is a scalable database that can be easily adapted for mobile, web and server development[17]. The latter will be used for the development of more complex data structures and provide responsiveness. Firebase Authentication will be in hand to allow secure user logins with robust features, the backend logic will be managed by Cloud Functions, such as running the matchmaking algorithm without the constraint of maintaining a server infrastructure. Finally, the Firebase SDK can be easily integrated into the Unity project.
- Blender: To create an immersive atmosphere in GlobeFair, Blender has been selected for the creation of high quality 3D assets, it's a powerful open source tool for 3D modelling, animation and rendering of object[18]. It's the prime candidate to shape the assets for the virtual career fair, such as avatars, booths, and environments elements, which then can be easily imported into Unity. Therefore, using this software aligns with the software commitment of delivering a high quality and realistic experience to the user base.

5.2 – Algorithms and Data Structures

- Matchmaking Algorithm: Will be implemented to allow straightforward, tag based matching systems of jobseekers and potential recruiters. The core idea of the algorithm is to match keywords or tags in both jobseekers CVs and the job descriptions. The recruiters/companies can add relevant tags to their listings for example required skills, job titles, and other custom attributes, while the jobseeker can personalise their profile to include their relevant skills, interests and experience if any. The system will work on a compatibility rate by providing the user with a score based on the number of common/matching tags.

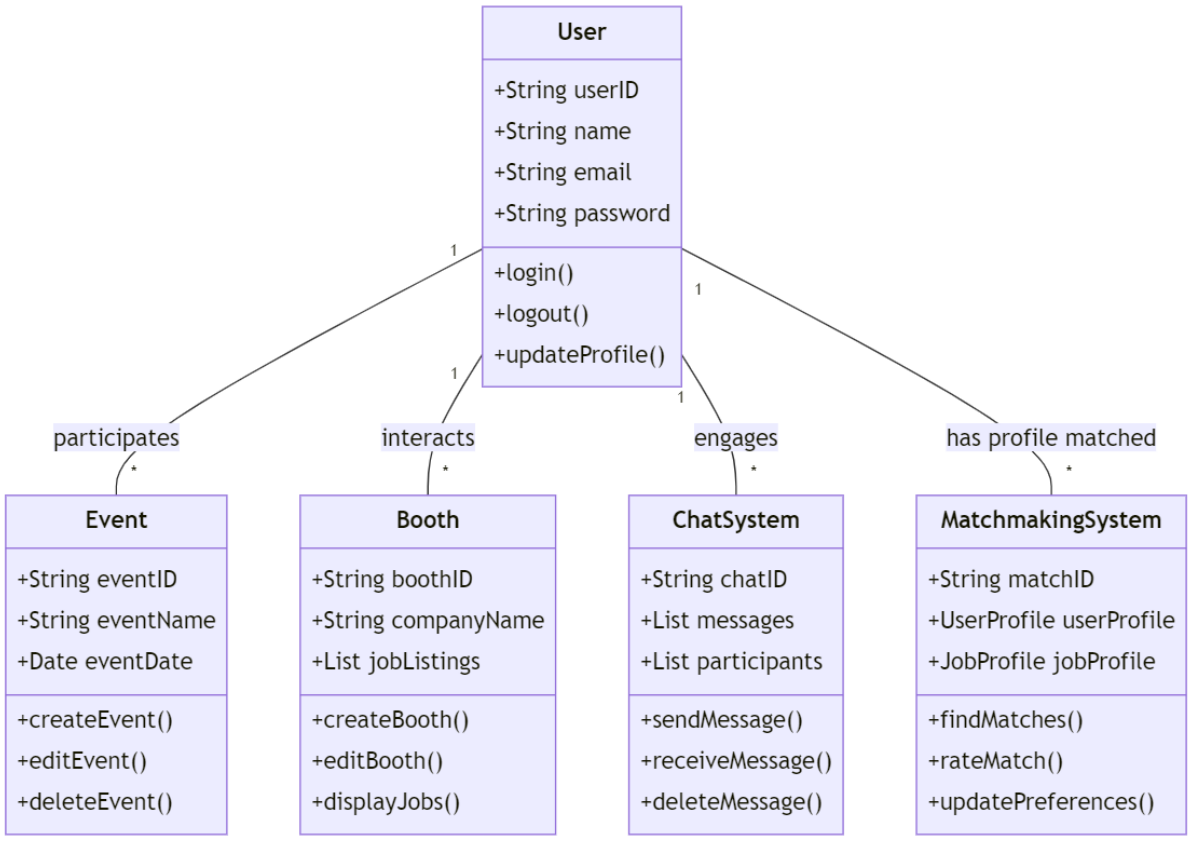
5.3 – Key Components

- API Integration: Google Cloud translation API is a crucial component of GlobeFair, it will allow to beat language barriers, by providing in real time translations of the chat and the conversation held in the career fairs, creating a more inclusive environment. This integration will work together with Firebase APIs which will enhance response time.

5.4 – Technical Details

- Responsive Design and IDE: Unity's User Interface toolkit will be fundamental to develop a responsive interface, which can be available on different user's devices, adapting to the diverse screen sizes. The coding process in the development of the software, will be held in the IDE, Microsoft Visual Studio, which is already integrated with unity provides a simple but yet powerful environment for writing and debugging C# scripts[20]. Figure 5.1, represents a class diagram, which allows to understand the software structure and how different parts of the system interact with each other.

Figure 5.1



6. Planning and Timescales

6.1 - Semester 1 Detailed Plan

Task	State	Start Date	Completion Date
Market and Competitor Analysis	Complete	08/09/2023	21/09/2023
Software Design and Feature Planning	Complete	21/09/2023	11/10/2023
Interim Report Writing	Complete	12/10/2023	23/11/2023
Aims & Objectives Identification	Complete	12/10/2023	20/10/2023
Requirements Outline	Complete	19/10/2023	25/10/2023
Business Model and Strategy	Complete	25/10/2023	07/11/2023
Design & Algorithms Outline	Complete	07/11/2023	20/11/2023
Interim Report Finalisation	Complete	20/11/2023	23/11/2023
Project Development Initialisation	Complete	24/11/2023	07/12/2023
Unity, IDE and Database Setup	Complete	24/11/2023	07/12/2023

6.2 - Semester 2 Sketch Plan

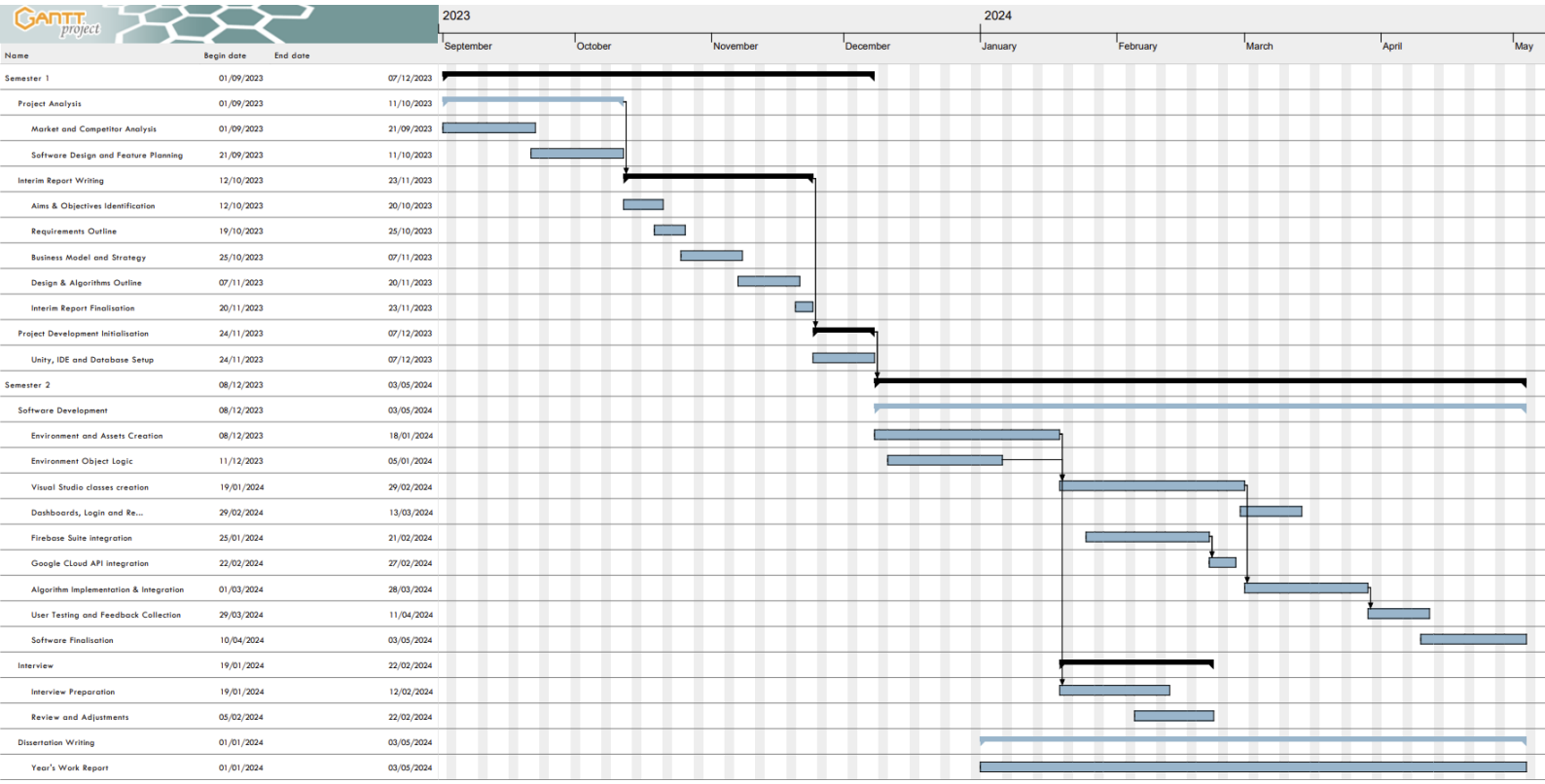
Task	State	Start Date	Completion Date
Software Development	Not Started	08/12/2023	03/05/2024
Environment and Assets Creation	Not Started	08/12/2023	18/01/2024

Environment Object Logic	Not Started	11/12/2023	05/01/2024
Visual Studio classes creation	Not Started	19/01/2024	29/02/2024
Dashboards, Login and Register Creation	Not Started	29/02/2024	13/03/2024
Firebase Suite integration	Not Started	25/01/2024	21/02/2024
Google CLoud API integration	Not Started	22/02/2024	27/02/2024
Algorithm Implementation & Integration	Not Started	01/03/2024	28/03/2024
User Testing and Feedback Collection	Not Started	12/04/2024	25/04/2024
Software Finalisation	Not Started	10/04/2024	03/05/2024
Interview	Not Started	01/02/2024	23/02/2024
Review and Adjustments	Not Started	05/02/2024	22/02/2024
Dissertation Writing	Not Started	01/01/2024	03/05/2024

6.3 - Gantt Chart Overview

The following Gantt chart provides a timeline for the development of the project. It outlines key tasks and milestones across the two semesters, ensures an organized strategy for the market research, software design, development, testing, and final deliverables.

Figure 6.1



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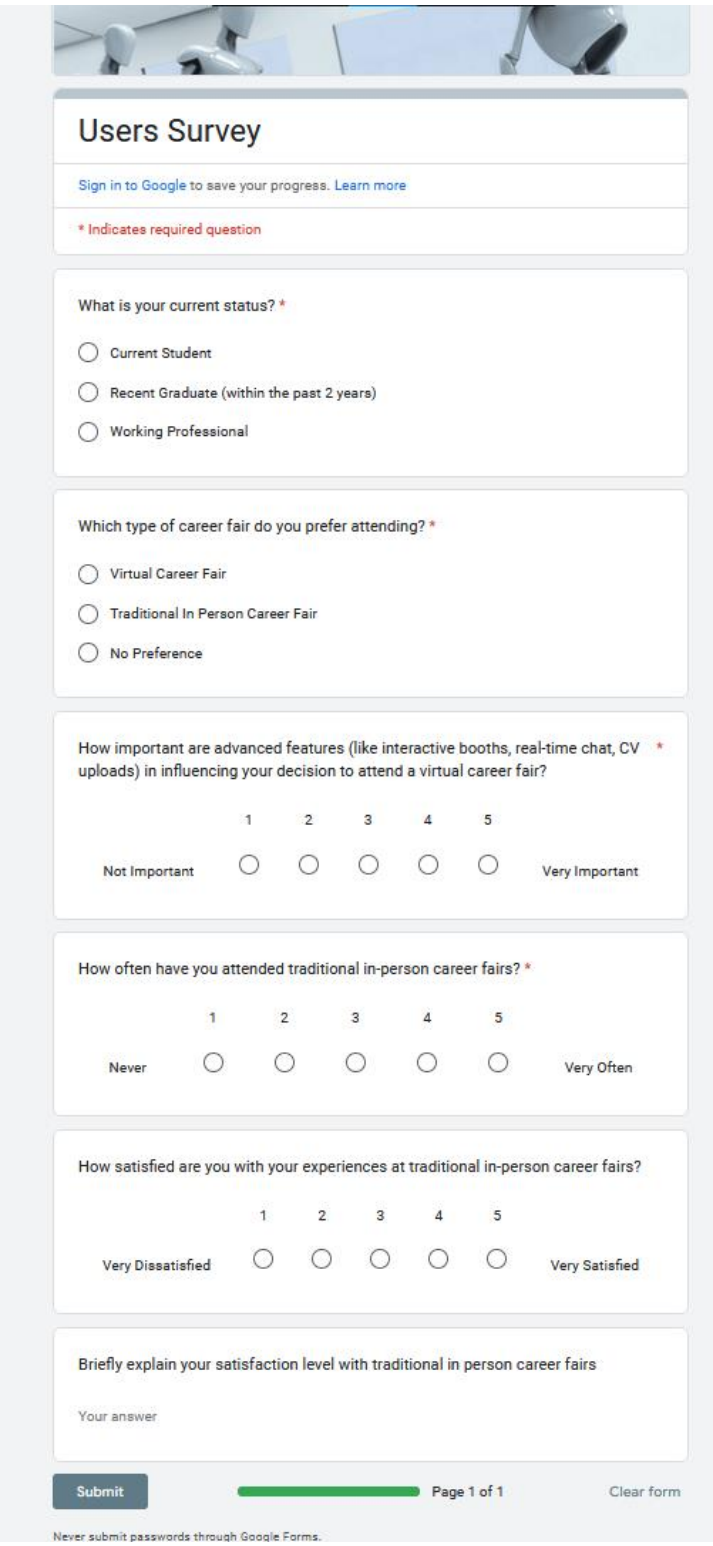
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APPENDIX 8

Figure 8.1



Users Survey

Sign in to Google to save your progress. Learn more

* Indicates required question

What is your current status? *

- Current Student
- Recent Graduate (within the past 2 years)
- Working Professional

Which type of career fair do you prefer attending? *

- Virtual Career Fair
- Traditional In Person Career Fair
- No Preference

How important are advanced features (like interactive booths, real-time chat, CV uploads) in influencing your decision to attend a virtual career fair? *

	1	2	3	4	5	
Not Important	○	○	○	○	○	Very Important

How often have you attended traditional in-person career fairs? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

How satisfied are you with your experiences at traditional in-person career fairs?

	1	2	3	4	5	
Very Dissatisfied	○	○	○	○	○	Very Satisfied

Briefly explain your satisfaction level with traditional in person career fairs

Your answer

Submit Page 1 of 1 Clear form

Never submit passwords through Google Forms.

Figure 8.2

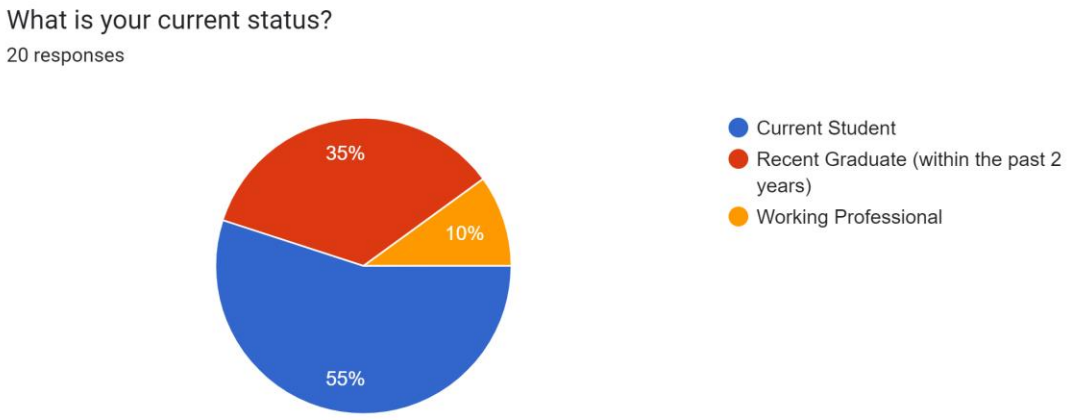


Figure 8.3

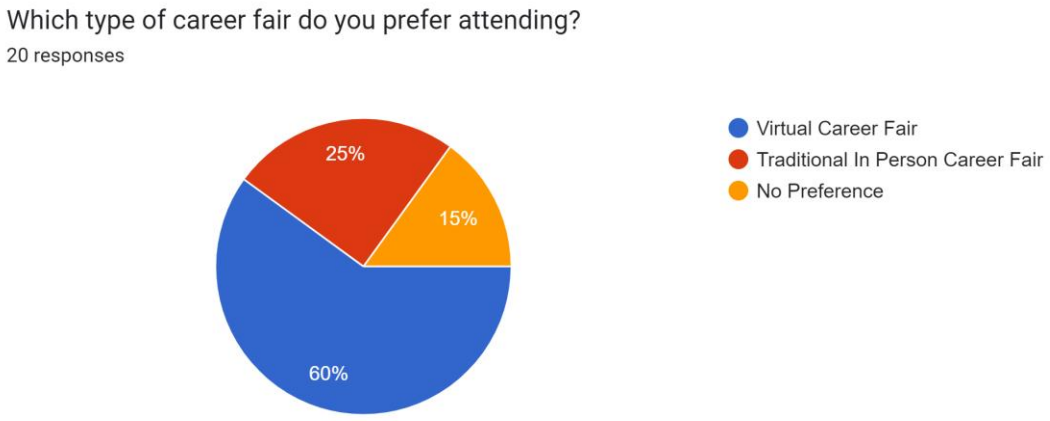


Figure 8.4

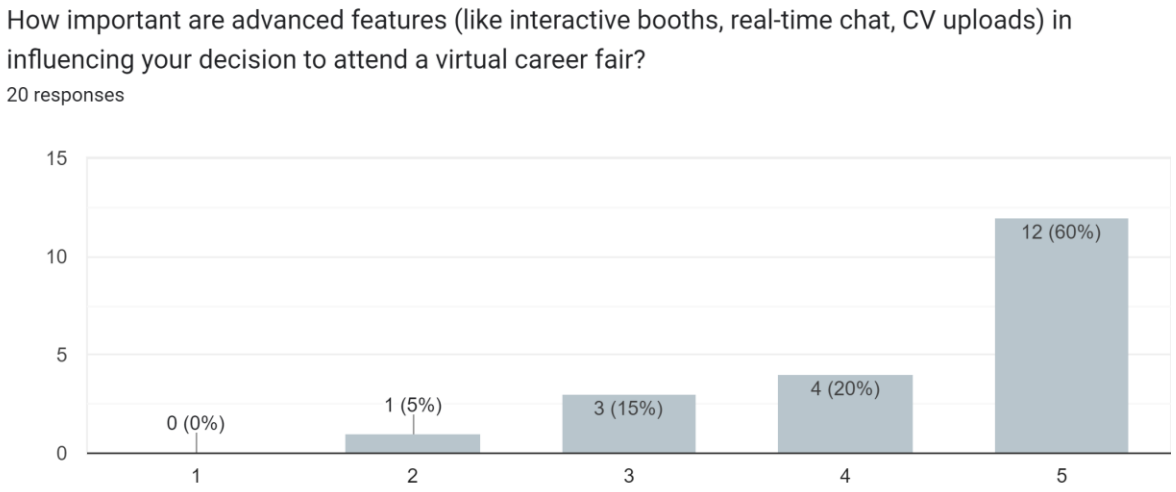


Figure 8.5



Figure 8.6

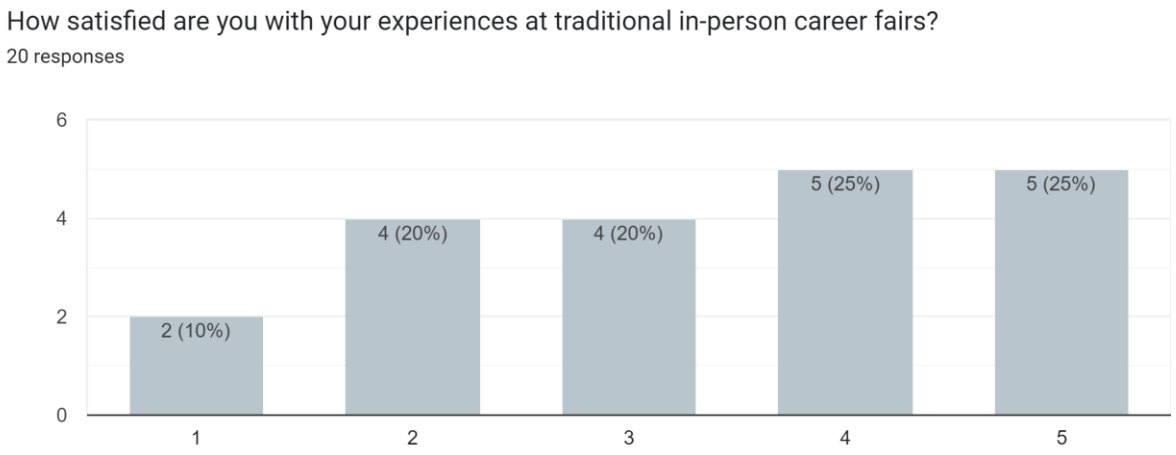


Figure 8.7

Briefly explain your satisfaction level with traditional in person career fairs

20 responses

The need of more engaging tools in traditional fairs
After the fair there are no follow up s
Because there arent enough directions to take full advantage of traditional career fairs
It offers a lot of insight into the companies and get to know different people
It could be improved but you gain a good idea of what the job requires in a traditional fair
Meet recruiters face to face
Attendees need more engaging activities
Good to get started into a career path
Don't really get helped or contacted after
They often lack guidance
Neutral feeling
It helps find companies you didn't hear about but need more interaction
Not attended much do to time constraints
Is only available during short periods of time and feels rushed
Should have more engaging activities
On the main note is good but dont get to learn about all companies
It's good for the idea but then everything is filled up online
You don't get to network or find new people
Have a direct conversation with people but no guidance afterwards
limited companies choice and way too many people